Mohammad Abdul Mohaimin

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Education

Bangladesh University of Engineering and Technology (BUET) Feb 2020 – Apr 2025
B.Sc. in Mechanical Engineering *Dhaka, Bangladesh*
Relevant coursework: Mechanics of Solids, Numerical Analysis, Metallic Materials, Composite Materials, Machine Design, Mechanics of Machinery
Research Interest: Computational Mechanics, Solid Mechanics, Mechanics of Materials, Composite Materials, Multiphysics Modeling, Numerical Modeling, Structural Integrity, Material Degradation and Damage Mechanics, Corrosion and Failure Analysis, Non-Destructive Testing, Welding and Materials Characterization

Research Experience

Multiphysics Modeling of Hybrid Laminated Composite Conductors Under a DC Field *Undergraduate Thesis — Supervisor: Dr. Shaikh Reaz Ahmed, Professor, Dept. of Mechanical Engineering, BUET*

- Developed a semi-analytical multiphysics framework for hybrid particle-reinforced laminated composite conductors subjected to a direct-current (DC) field, capturing the coupled electro-thermal-mechanical response.
- Formulated the mechanical problem using a Displacement Function approach, reducing the coupled two-dimensional mechanical equilibrium equations to a single fourth-order partial differential equation and enabling a more efficient numerical formulation.
- Implemented a Finite Difference Method (FDM) with iterative solvers to solve the coupled electro-thermal-mechanical system, incorporating nonlinear Joule heating and temperature-dependent electrical resistivity and thermal conductivity.
- Applied the model to Al-SiC and Al-Graphene symmetric laminates with continuous power-law volume-fraction gradation and intermediate bonding layers, and validated voltage, temperature, and stress predictions against COMSOL Multiphysics simulations.

Research Publications & Presentations (In progress)

- "Electro-thermo-mechanical coupling in functionally-graded composite laminates: A Displacement-Function approach" — M. S. W. Ullah, T. Islam, **M. A. Mohaimin**, M. N. Shahriare, S. R. Ahmed — ***Manuscript in preparation.***
- "Assessment of Landfill Gas-to-Energy Potential and Technologies at Arefin Nagar and Fatehabad Landfills Using Site-Specific LandGEM Modeling" — P. Biswangri, A. Chowdhury, A. Mandal, **M. A. Mohaimin** — *Accepted for presentation (In Dec 2026)*, 8th International Conference on Chemical Engineering (ICCHE), BUET, 2026.

Academic Engineering Projects

Design, Fabrication, and CFD Analysis of a Double-Pipe Heat Exchanger with Disc-Type Turbulator and Outside Fins

- Designed and fabricated a 1-metre counter-flow double-pipe heat exchanger incorporating a 13-disc perforated turbulator and 8 external longitudinal fins for passive heat-transfer enhancement.
- Performed MATLAB-based NTU-effectiveness analysis and 3D CFD analysis in SimScale using a hexahedral mesh, demonstrating an additional 2 °C temperature drop due to the turbulator.
- Verified the structural and pressure-boundary integrity of the mild-steel prototype through a 400 psi hydrostatic pressure test.

Mitigation and Utilization of Methane Emissions from Landfills of Dhaka: Waste-to-Energy Feasibility Study

- Applied Landfill Gas Emissions Model to estimate methane generation at Matuail and Aminbazar landfills through 2045 using Bangladesh-specific parameters.
- Evaluated electricity-generation potential through anaerobic digestion and incineration and assessed greenhouse-gas mitigation through landfill-gas recovery.

Autonomous Detection of Potential Mosquito Breeding Sites Using UAVs and Computer Vision

- Designed, fabricated, and field-tested a custom quadcopter UAV integrating an onboard camera, GPS telemetry, flight controller, BLDC motors, and telemetry systems.
- Trained and implemented a YOLOv5 object-detection model on a custom dataset to identify stagnant water and potential mosquito breeding sites from aerial imagery.

Professional Experience

Bashundhara Oil & Gas Company Ltd. Jan 2026 – Present
Assistant Engineer — Plant Inspection & Integrity

- Oversee the mechanical integrity and reliability of 250+ static-equipment assets, including pressure vessels, heat exchangers, storage tanks, and process piping, ensuring full compliance with API, ASME, and other internationally recognized codes and standards.
- Lead inspection planning, damage-mechanism identification, failure-mode analysis, and root-cause analysis, translating corrosion-rate and remaining-life findings into repair, replacement, and design-modification recommendations that address material degradation and structural stability under demanding thermal, mechanical, and chemically aggressive operating conditions.
- Verify material quality and oversee NDT, pressure-testing, welding, and coating activities while leading a team of four inspection specialists and coordinating with production, maintenance, planning, and EHS departments.

Abul Khair Group May 2025 – Dec 2025
Technical Management Trainee — Service Management

- Led technical service operations for 7,500+ coffee vending machines and ∼42 service officers across one third of Bangladesh; developed data-driven service-quality reporting and coordinated technical support and spare-parts procurement.

Technical Skills

Computational & Numerical: MATLAB, Python, C, Finite Difference Methods, Numerical Analysis, Multiphysics Modeling
Simulation: COMSOL Multiphysics, ANSYS Workbench, Abaqus, SimScale, SolidWorks, HTRI, AutoCAD, LaTeX
Engineering: Mechanical Integrity, NDT (UT, PT, MT, VT), Corrosion and Degradation Assessment, Remaining-Life Evaluation, Root Cause Analysis, Welding Inspection

Certifications

- **ISO 9001:2015 QMS & ISO 14001:2015 EMS Internal Auditor** (BOGCL, 2026)
- **Pressure Vessel Design, Fabrication, Inspection & Testing per ASME BPVC** (BUET DCE)